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DEPARTMENT OF CIVIL, ENVIRONMENTAL
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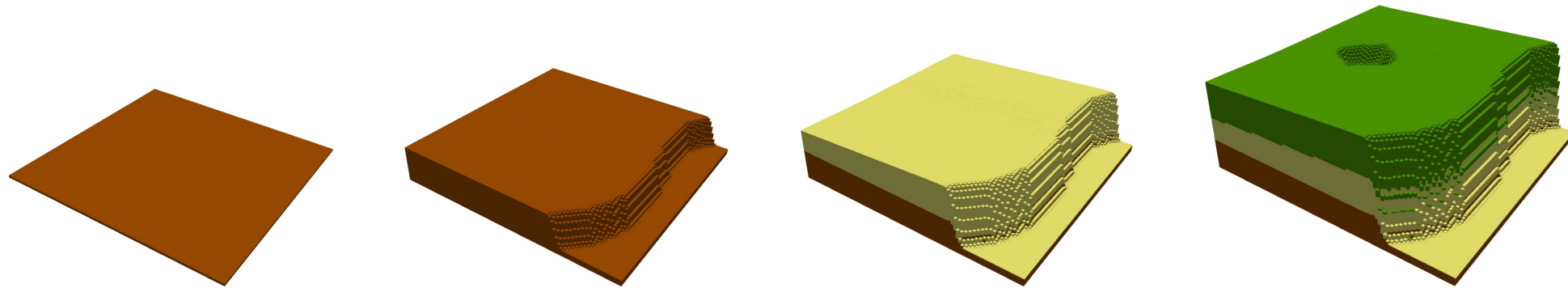
Numerical simulation of long-term accreting-compacting environments by a novel Finite Volume approach on a growing grid

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*Session 17 – Model coupling, domain decomposition, and solvers for multiphysics
problems*

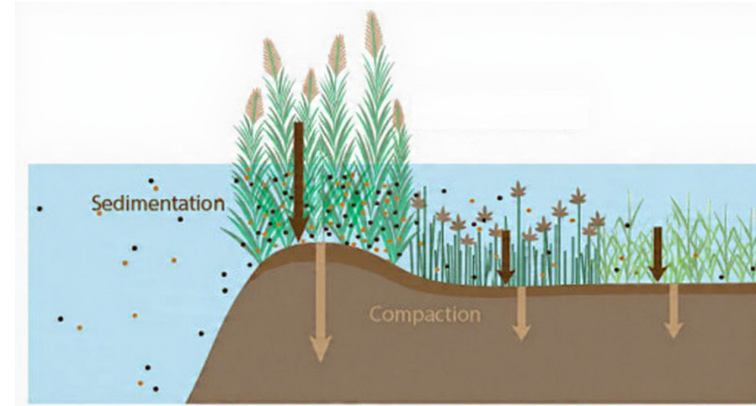
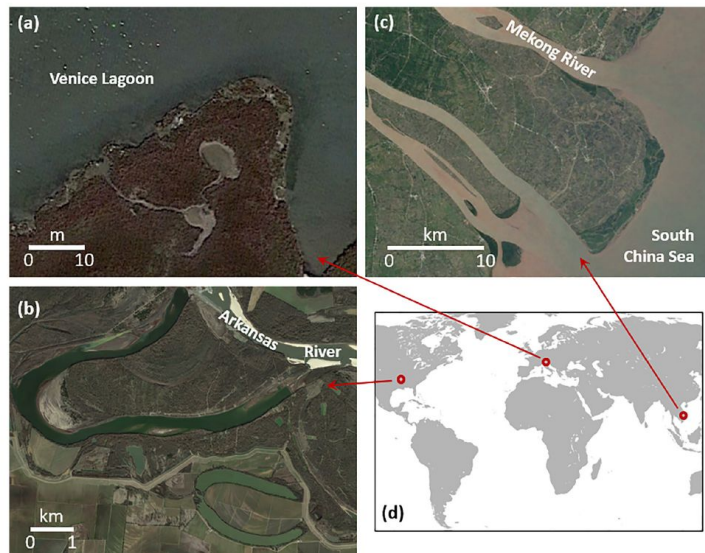
June 30, 2026

- **MOTIVATION** - physical phenomenal and current state of the art.
- **MATHEMATICAL AND NUMERICAL MODEL** - governing equations and discretization.
- **VALIDATION** - comparison with state of the art simulator.
- **MARSH EVOLUTION** - experiment on a real world case.
- **CONCLUSION**

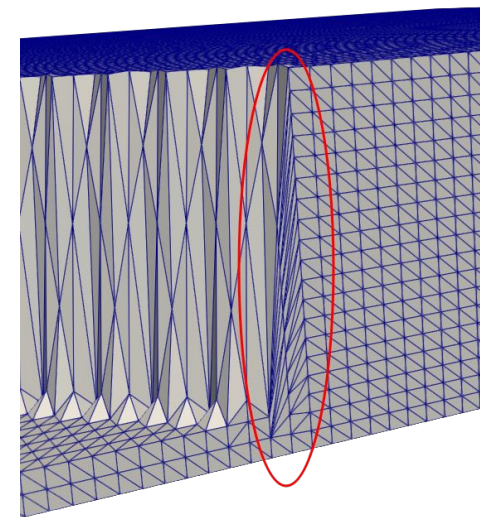


Coastal regions, wetlands, marshes, mangroves, and river deltas are dynamic environments shaped by sediment movement, water flow, and ecological processes.

- **Found worldwide**, they represent an important ecological and socio-economic region.
- They are **highly vulnerable** to sea-level rise, flooding, and human impacts.
- Understanding how these systems evolve is essential for **assessing risks** and predict their behavior.



These environments are typically composed of unconsolidated, water-saturated sediments that are **highly susceptible to self-compaction** and land subsidence.



One way to model landform evolution is with the finite element method (FEM); however, it can produce **highly distorted meshes**.

The experiments presented in [1] is used to validate the proposed method.

[1] Xotta, R., Zoccarato, C., Minderhoud, P. S. J., & Teatini, P. (2022). Modeling the role of compaction in the three-dimensional evolution of depositional environments. *Journal of Geophysical Research: Earth Surface*, 127, e2022JF006590. <https://doi.org/10.1029/2022JF006590>

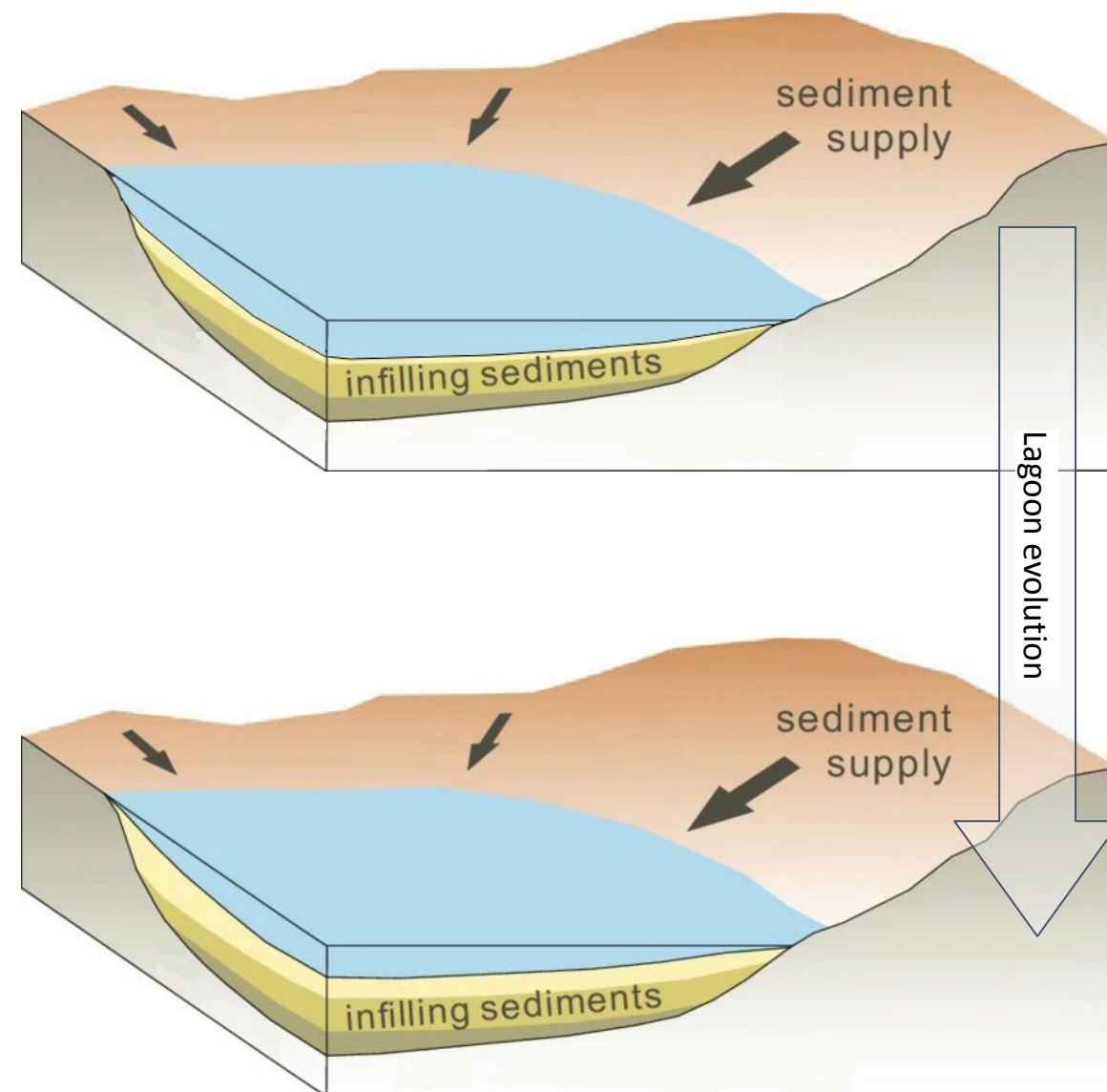
To study how sediment deposition affects the evolution of depositional environments, a mathematical model was developed with:

- A **hydrodynamic module** for single-phase flow
- A **geomechanical compaction module** based on Terzaghi's consolidation theory

Together, these modules quantify **natural compaction** and its influence on **landscape evolution**.

Main Features

- Simulates continuous sediment accumulation and landform growth
- Dynamically evolving domain
- Captures long-term natural subsidence processes



The system is written to find $\mathbf{p}$ in:

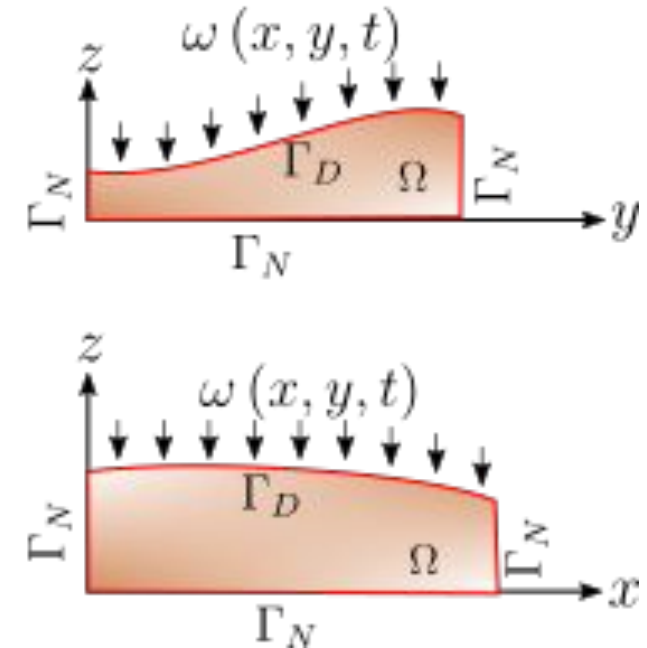
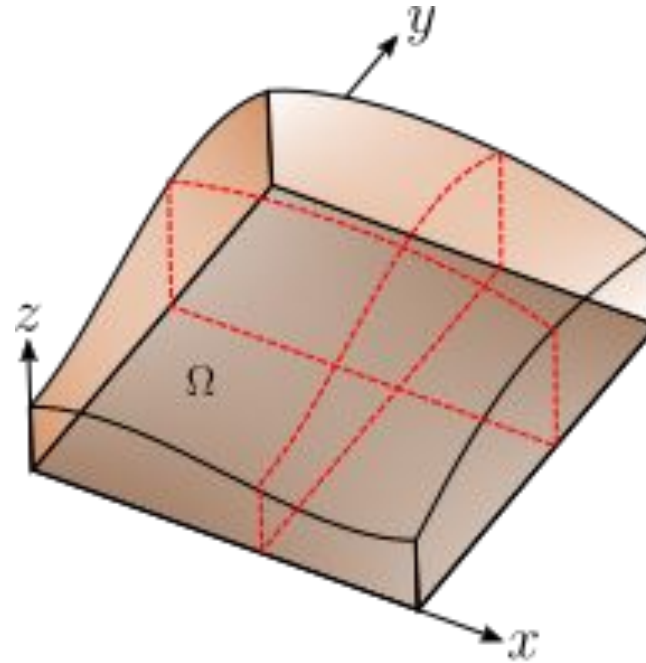
Mass Balance $\left(c_b + \frac{e\beta}{1+e}\right) \frac{\partial p}{\partial t} - \nabla \cdot \frac{\mathbf{K}}{\gamma} \nabla p = c_b \dot{\sigma}_z \quad \text{on } \Omega \times I$

Terzaghi's Principle $\hat{\sigma}_z = \sigma_z + p \quad \text{on } \Omega \times I$

Total stress Variation $\dot{\sigma}_z = (1 - \phi_0)(\gamma - \gamma_w)\omega \quad \text{on } \Omega \times I$

Oedometric Compressibility $c_b = \frac{1}{\ln 10} \frac{C_c}{(1+e)\sigma_z}$

Boundary Condition $-\frac{\mathbf{K}}{\gamma} \nabla p \cdot \mathbf{n} = 0, \quad \text{on } \Gamma_N \times I$
 $p = 0 \quad \text{on } \Gamma_D \times I$

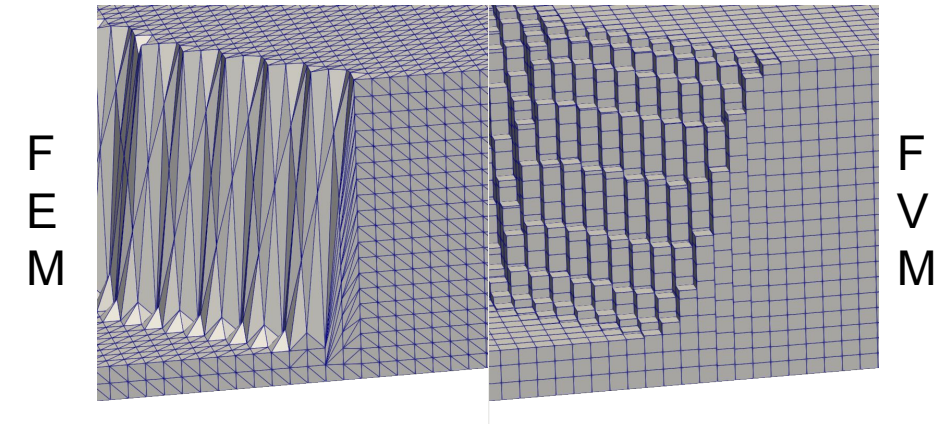


Constitutive Law
$$e = \begin{cases} e_0 - C_r \log \left(\frac{\sigma_z}{\sigma_{z,0}} \right), & \sigma_z < \sigma_{z,c} \\ e_0 - C_c \log \left(\frac{\sigma_z}{\sigma_{z,c}} \right), & \sigma_z \geq \sigma_{z,c} \end{cases}$$

The domain Ω evolves over the time interval I , expanding as sediment is deposited at a rate ω

Motivation

- The physical domain **continuously evolves** as sediment is deposited.
- Domain growth requires periodic **grid re-discretization**.
- Re-discretization introduces additional computational cost, which depends on the chosen discretization approach.
- The finite element method (**FEM**) tends to **create a distorted mesh** in areas where a natural steep embankment develops.

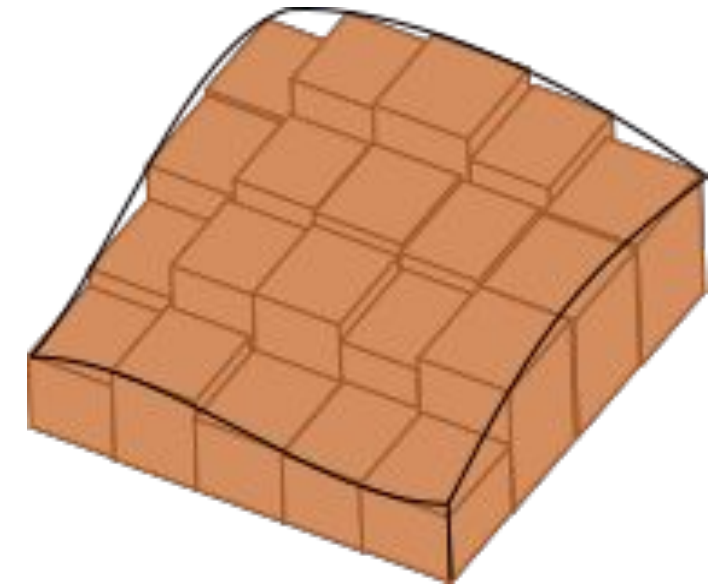


Proposed Approach

- Use a Finite Volume Method (**FVM**) on a **structured grid** to reduce the impact of domain growth on computational performance.

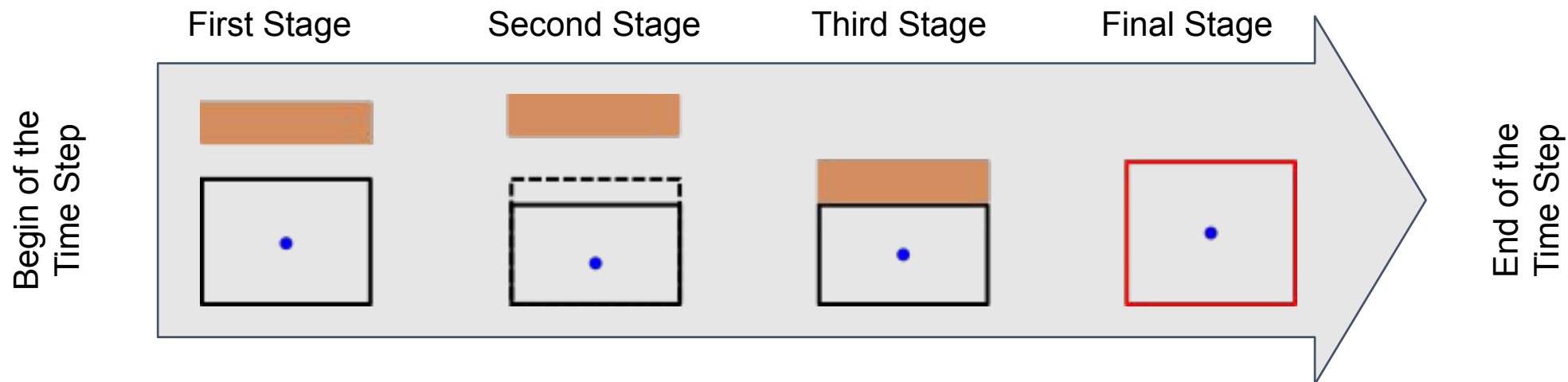
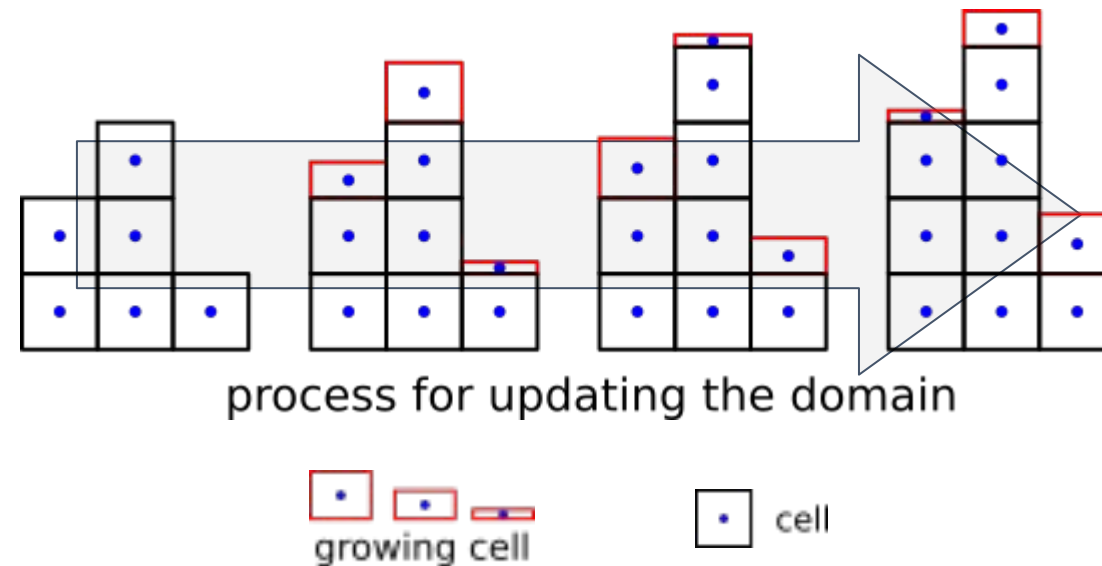
Main Advantages

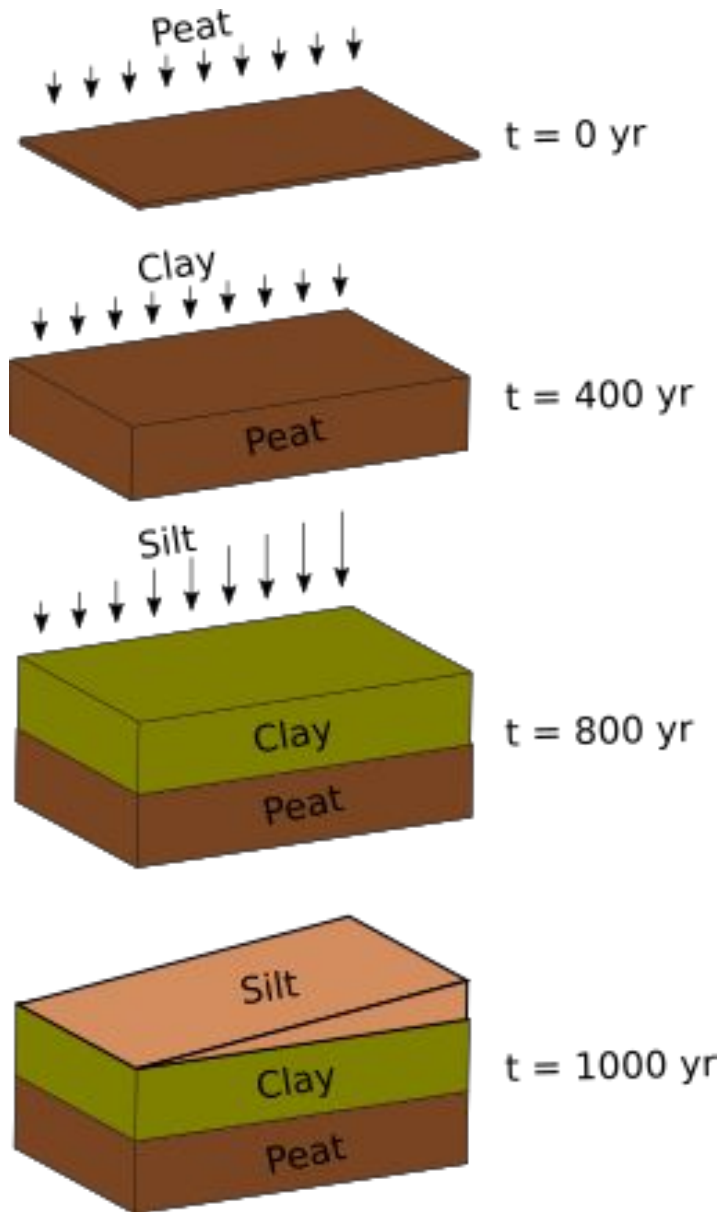
- **FVM**: The description is in the cell connectivity, this simplify the model deformation.
- **Stress computation**: the structure grid naturally align with the 1D Terzaghi's model.
- **Domain growth**: implemented by adding new cells at the top of older ones.



The discretization is **compose** by two kind of cell, one standard **cell** and a **growing cell**.

The growing cell was developed to **grow** in vertical size as function of the **sedimentation rate**, and is present only forming the surface at the top of the domain.





The experiment simulation represent a period of 1,000 years with three period of constant depositional, as :

- **0–400 yr:** Uniform spatial distribution of peat.
- **400–800 yr:** Uniform spatial distribution of clay.
- **800–1000 yr:** Linear spatial distribution of silt.

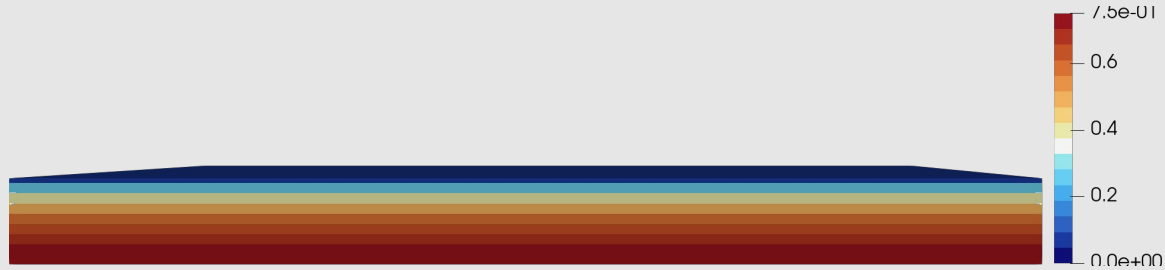
Table 1

Hydro-Geomechanical Properties of Typical Sediment Classes Composing Depositional Landforms Used in This Work

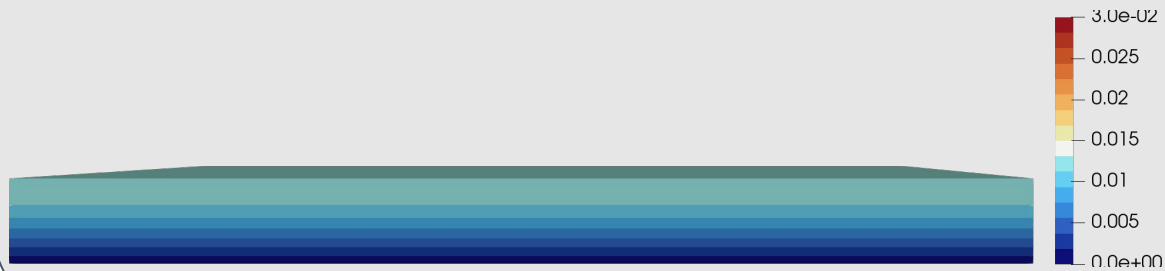
Material ID	$\sigma_{z,pre}$ (kPa)	γ_s (kN/m ³)	e_0 (–)	C_c (–)	C_r (–)	K_z (m/s)	$K_x/K_z, K_y/K_z$ (–)
Peat	1.0	20	15	4.0	0.40	10^{-7}	3.0
Clay	1.0	27	10	3.0	0.10	10^{-9}	3.0
Silt	1.0	27	3.0	0.5	0.05	10^{-7}	3.0
Sand	1.0	27	1.0	0.2	0.02	10^{-6}	3.0

Note. The values were derived after Brain et al. (2015), Bridgeman (2018), and Zoccarato et al. (2018). e_0 represents void ratio at $\sigma_z = \sigma_{z0} = 0.01$ kPa.

FVM

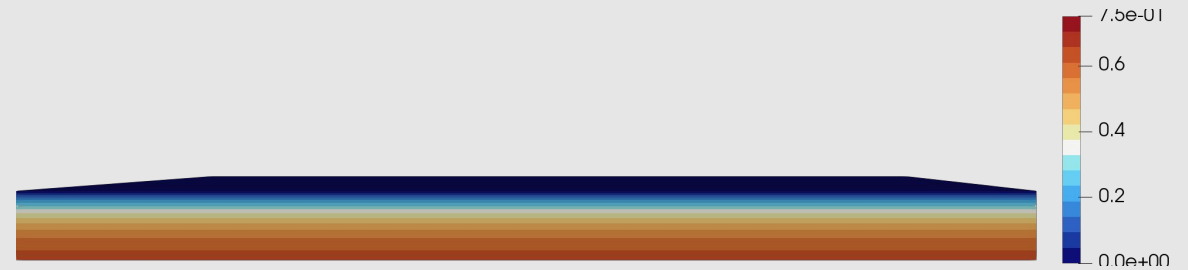


Pressure (Pa)

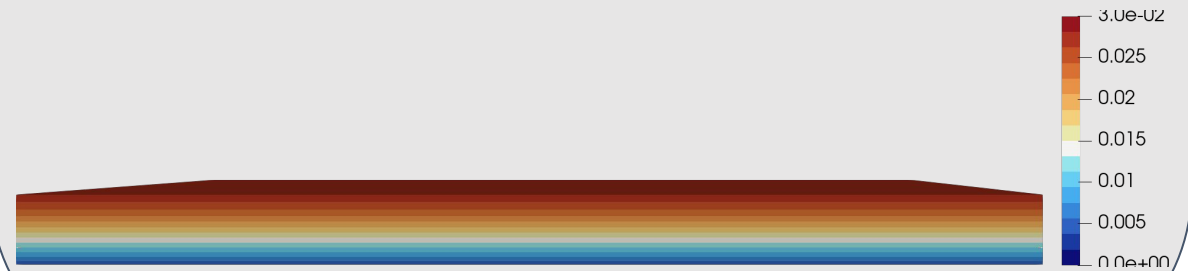


Compaction (m)

FEM

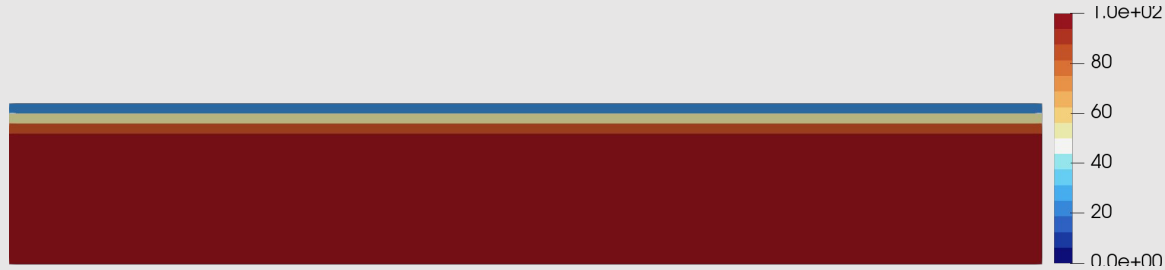


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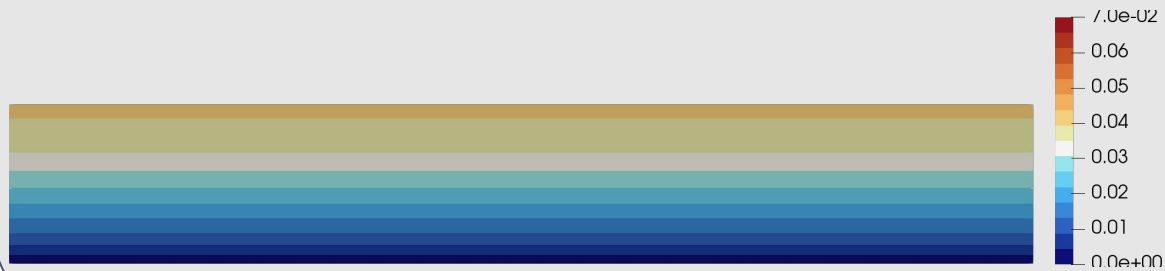


Compaction (m)

FVM



Pressure (Pa)

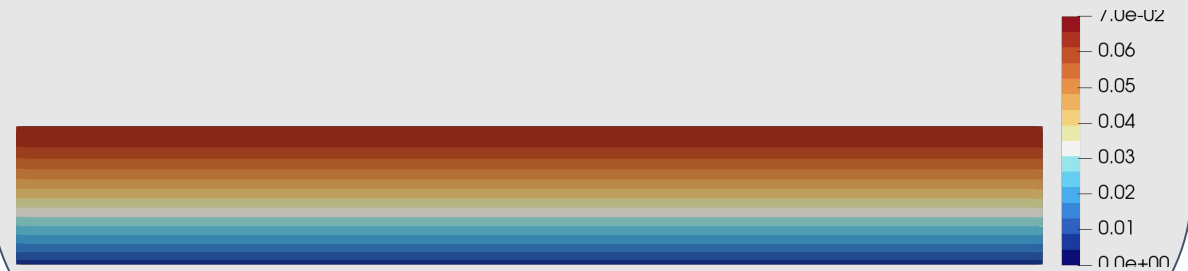


Compaction (m)

FEM

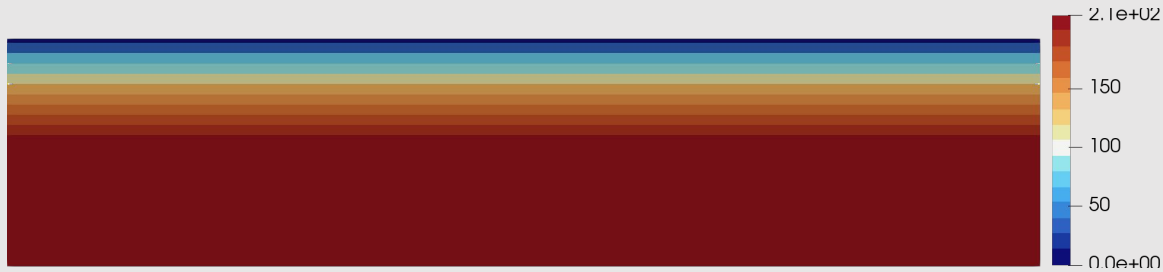


Pressure (Pa)

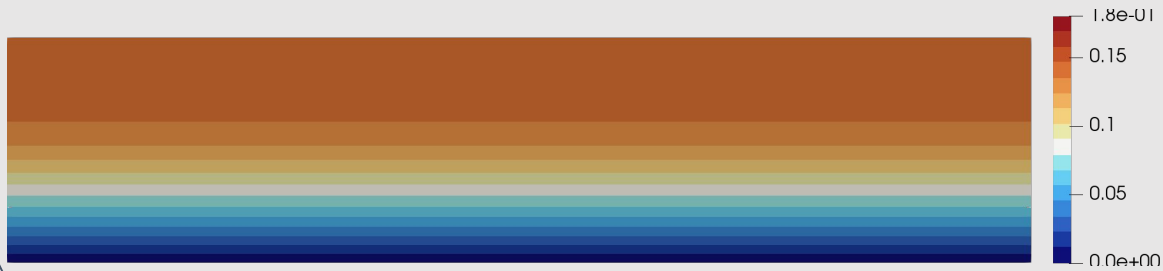


Compaction (m)

FVM

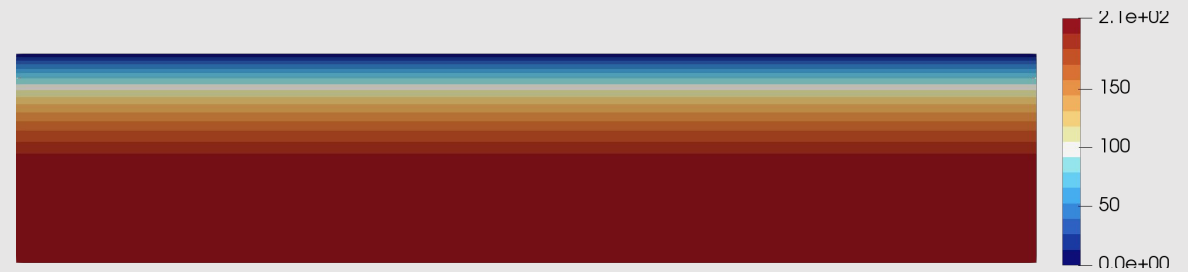


Pressure (Pa)

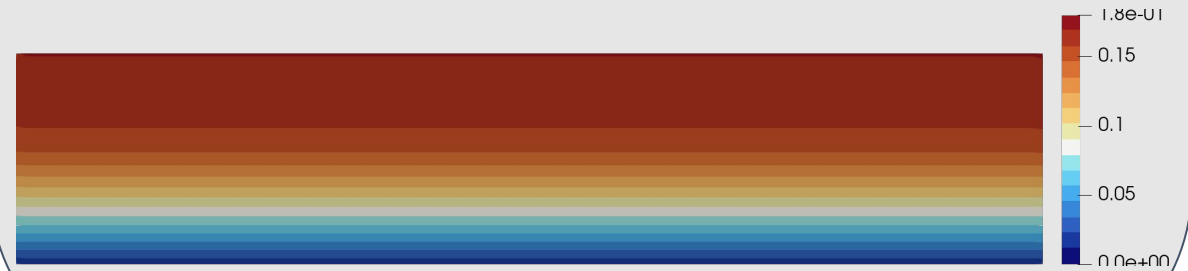


Compaction (m)

FEM

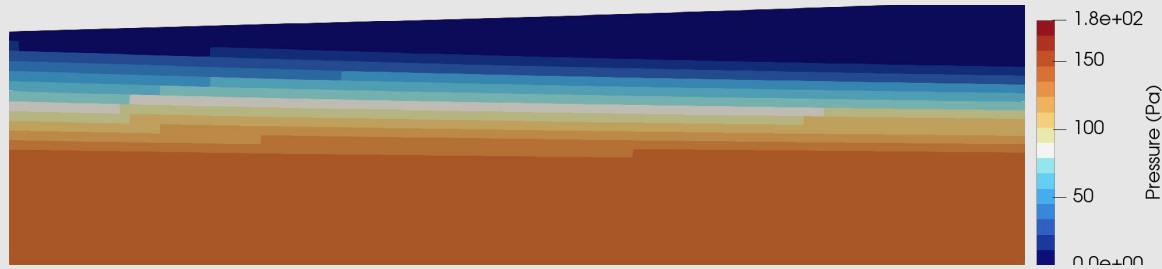


Pressure (Pa)

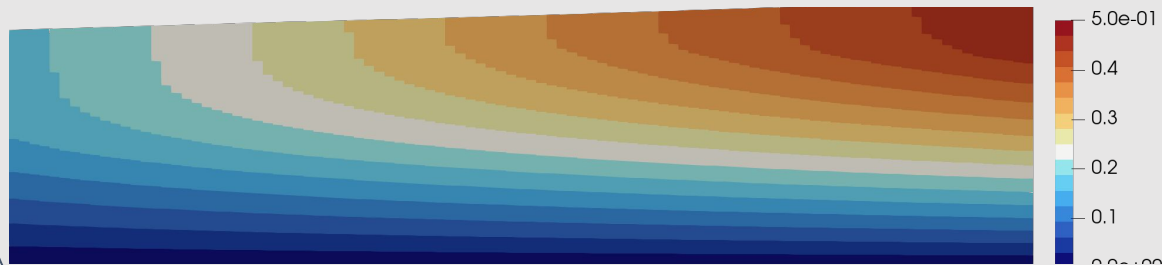


Compaction (m)

FVM

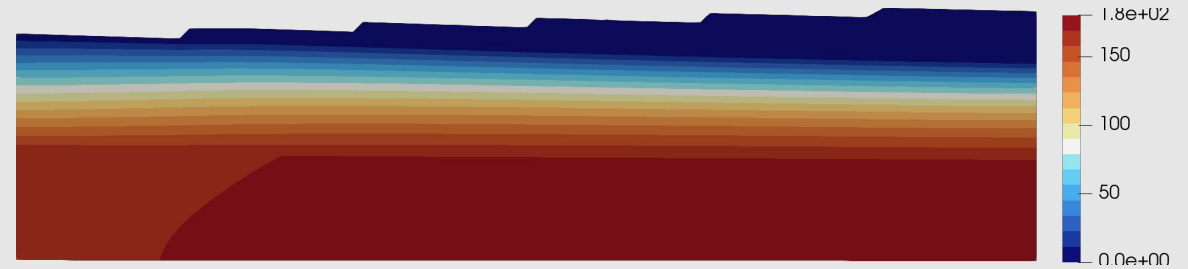


Pressure (Pa)

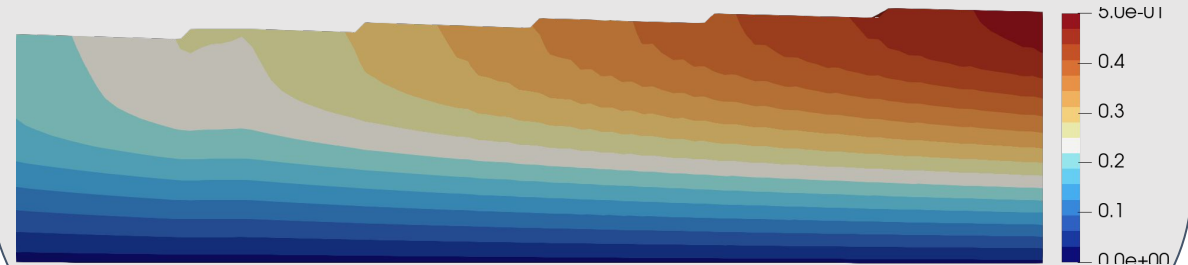


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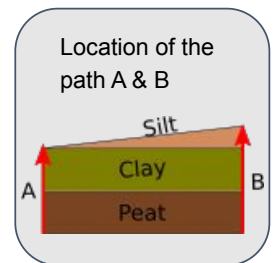
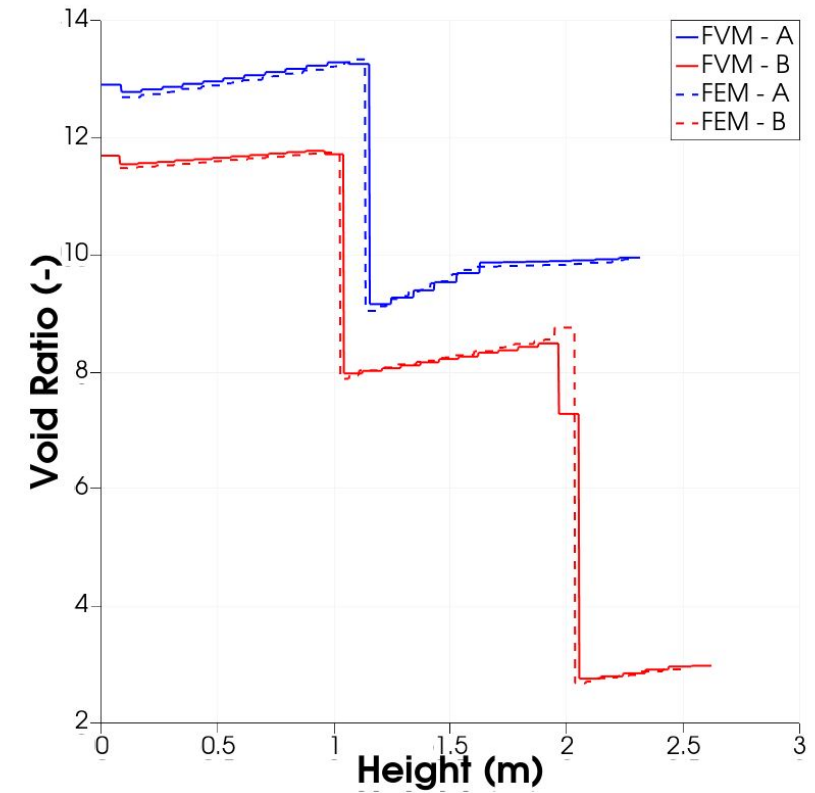
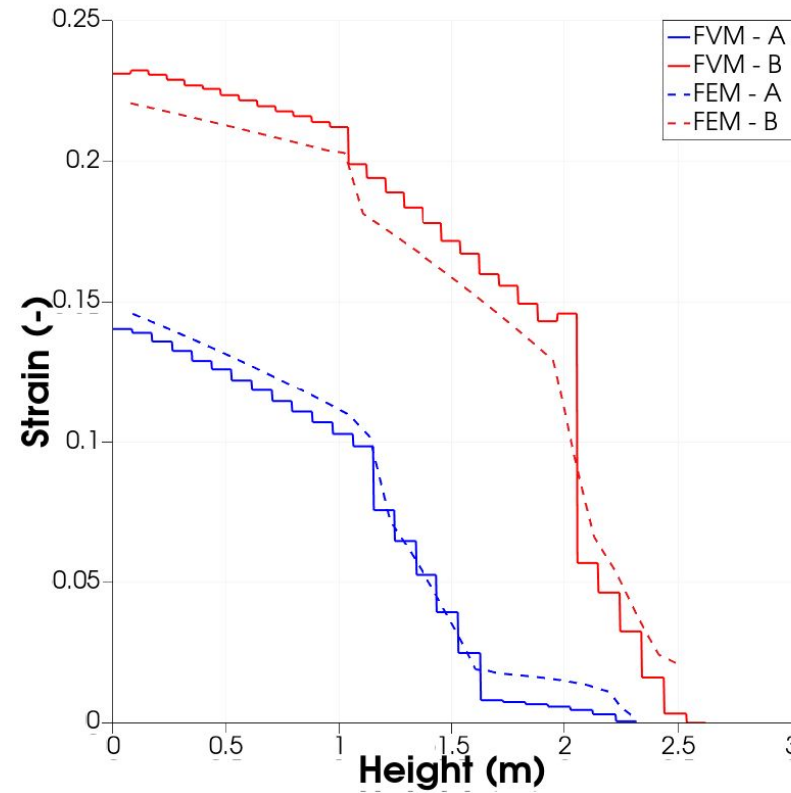
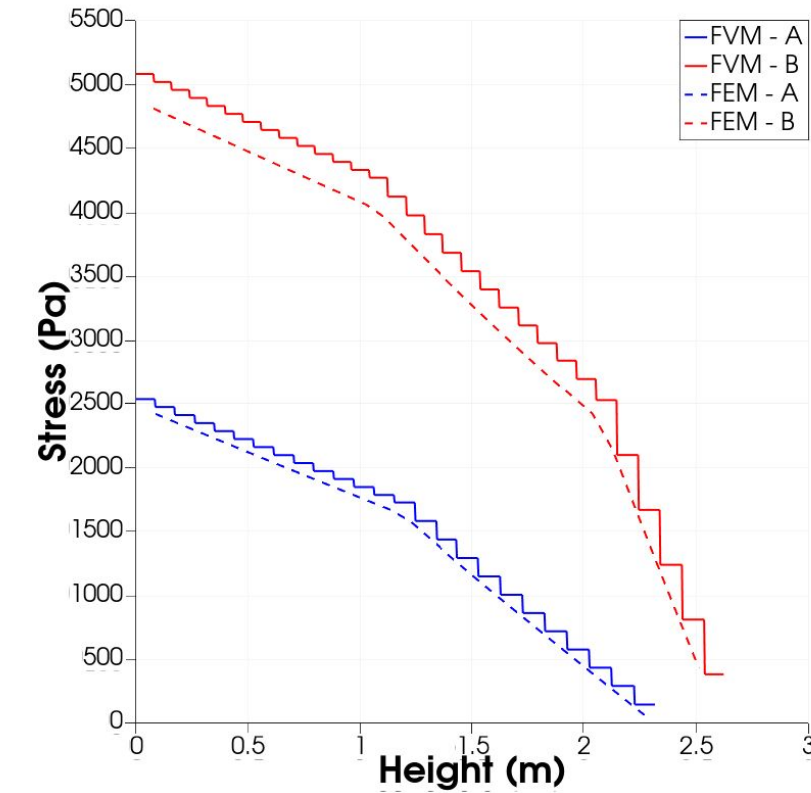
FEM



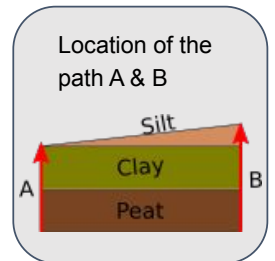
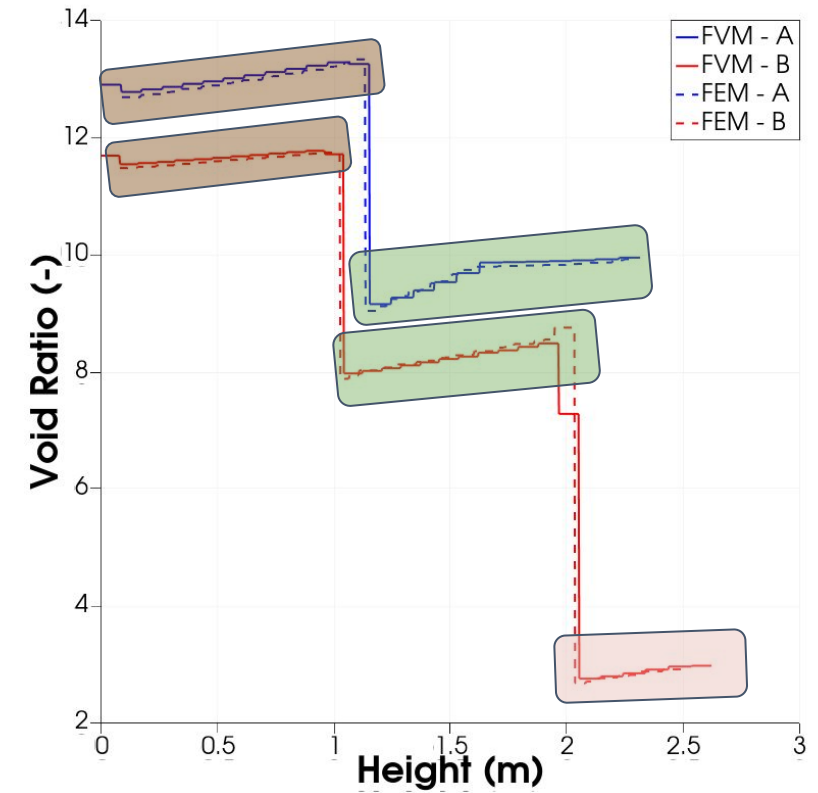
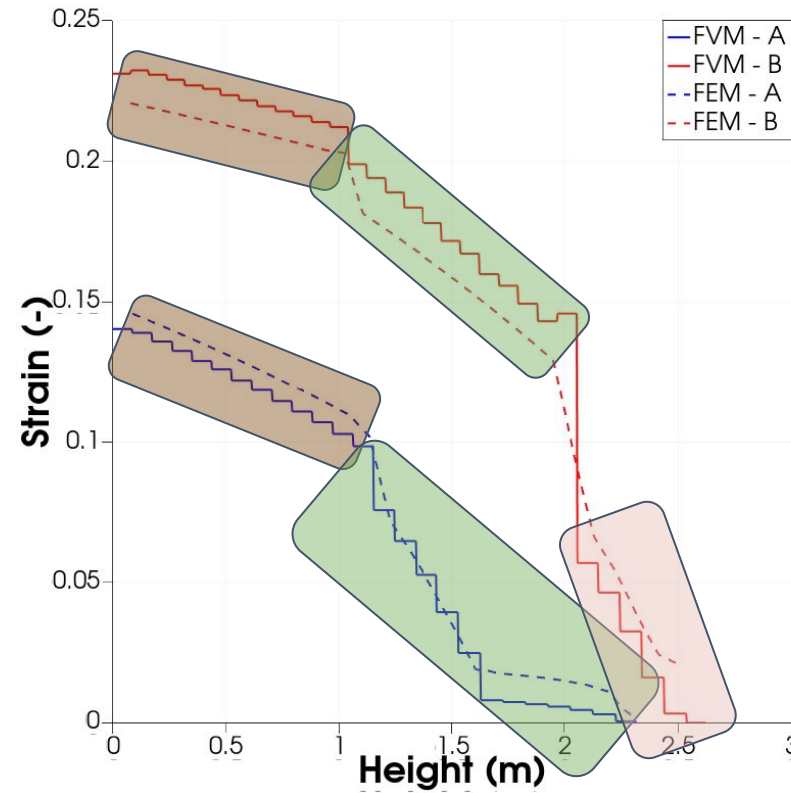
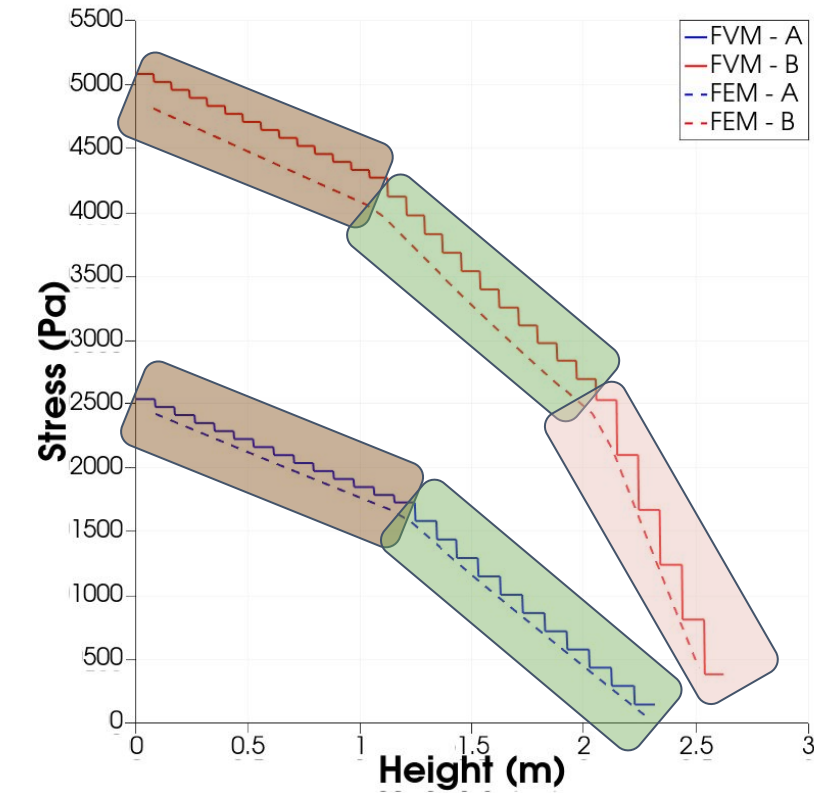
Pressure (Pa)



Compaction (m)

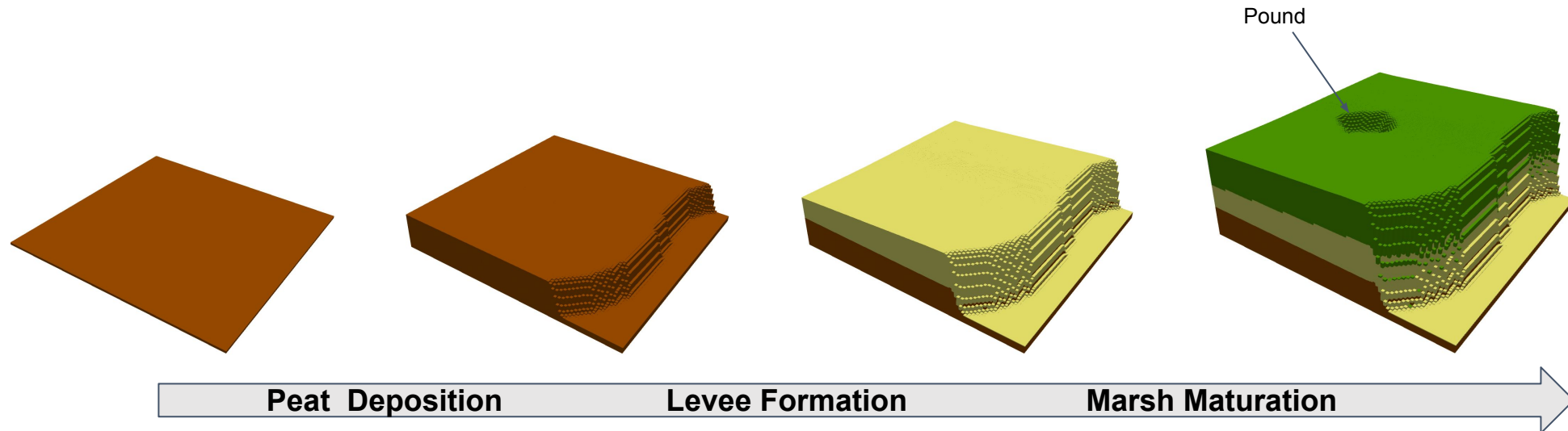


Numerical results - time = 1000 years

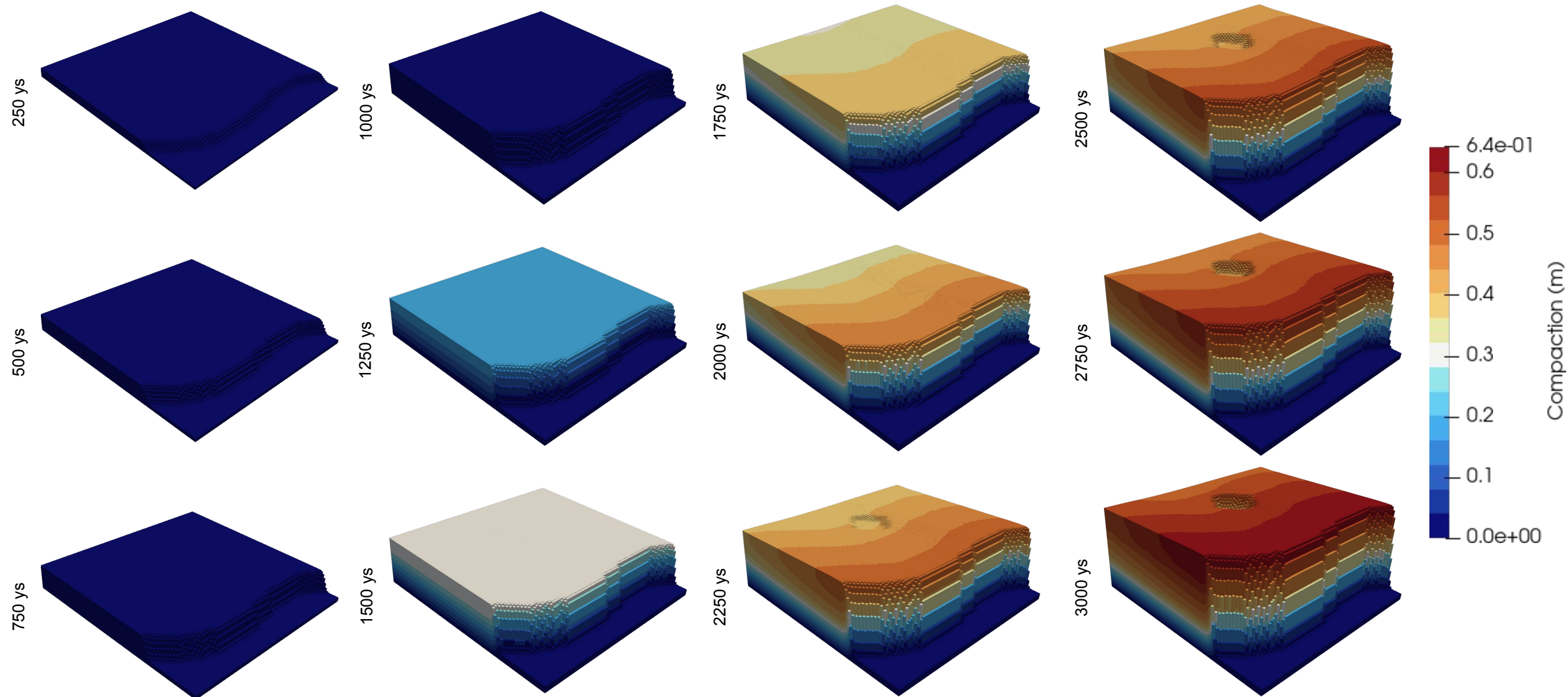


Consist of a spans over a period of 3,000 years and considers three main depositional phases:

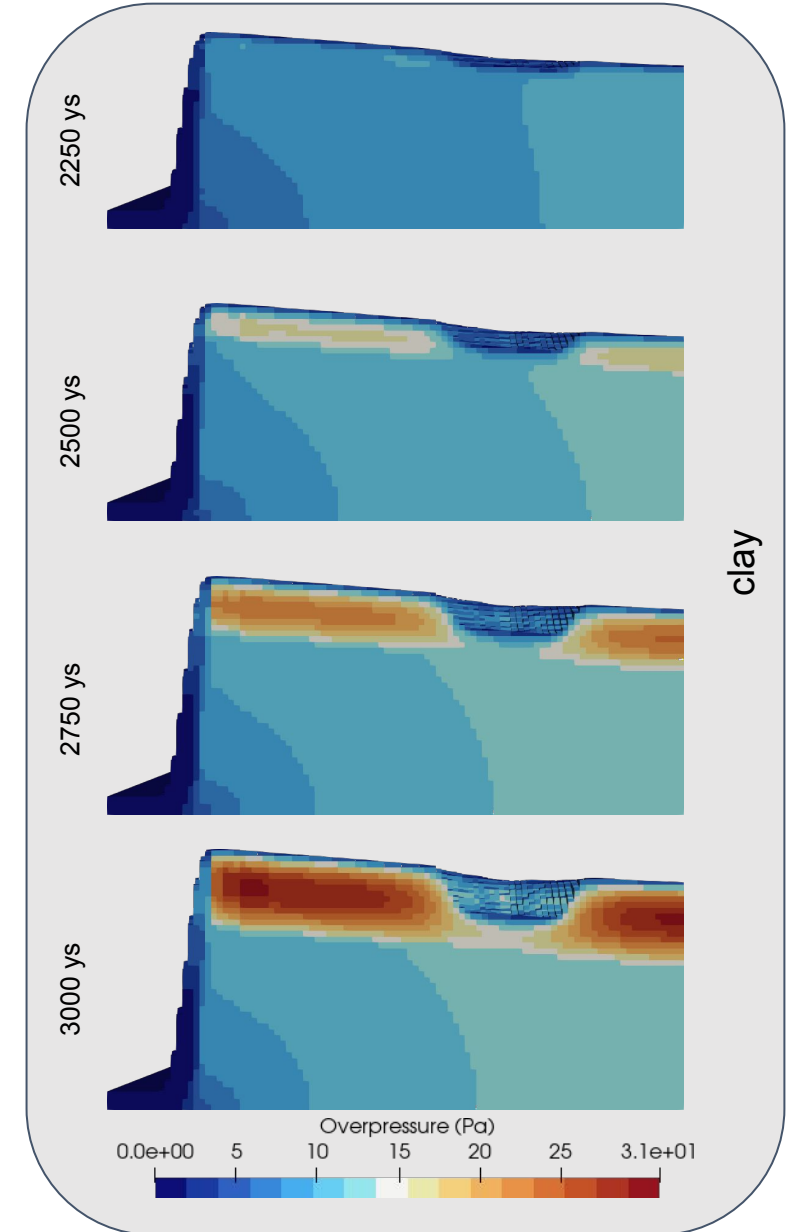
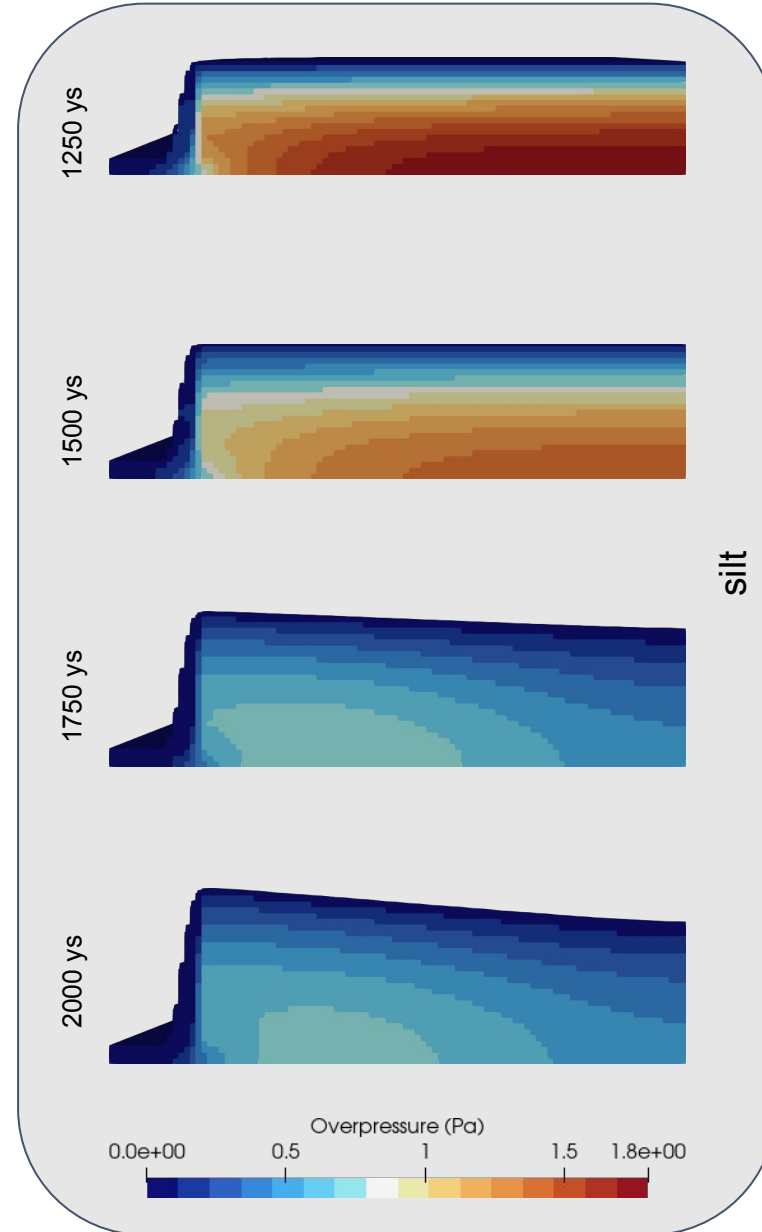
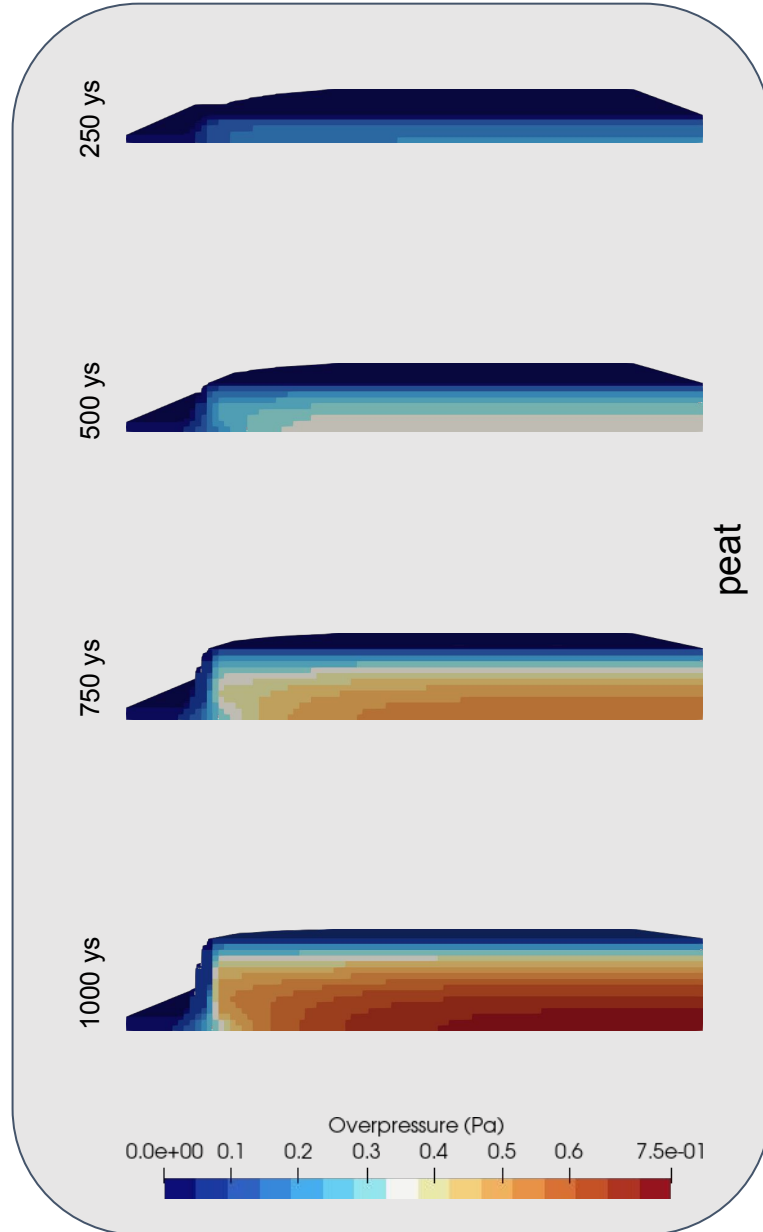
- **Peat accumulation phase (0–1000 yr)**
- **Overbank and levee formation phase (1000–2000 yr)**
- **Salt marsh maturation phase (2000–3000 yr)**



Numerical results - Evolution of the compaction



Numerical results - Evolution of the pressure field



- ❖ The growing-cell approach improves the surface resolution at the top of the domain, providing a **more accurate representation of smooth surfaces**.
- ❖ The proposed discretization **significantly simplifies the grid reconstruction process**. Existing cell connections are preserved, and only the connections associated with newly created cells need to be added.
- ❖ The proposed discretization produces **results that are consistent** with those reported in the literature.
- ❖ Unlike FEM, the FVM formulation **avoids mesh distortion** and is better suited to describe natural steep embankment.

For future improvements:

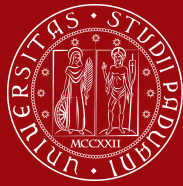
- ❖ Apply the methodology to complex sedimentary environments, such as the Venice Lagoon and the Mekong Delta.
- ❖ Incorporate creep behavior into the constitutive law.
- ❖ Extend the model to account for erosion processes.

Everything is available in:

GReS

Geomechanical Reservoir Simulator





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**Thank you for your attention
Questions?**